4
Applications of the Euclidean Algorithm

This chapter presents several applications of the Extended Euclidean Algorithm: modular arithmetic, in particular modular inverses; linear Diophantine equations; and continued fractions. The latter in turn are useful for problems outside of computer algebra: devising astronomical calendars and musical scale systems.

4.1. Modular arithmetic

We start with some applications. The first one is checking programs for correctness. In Part II of this book, we will see extremely fast algorithms for multiplication of large integers. These methods are also considerably more complicated than classical multiplication, and an implementation quite error-prone. So we may want to test correctness on many inputs. We take inputs a and b, say positive integers of 10000 words each, and the output c of 20000 words. Can we check that $a \cdot b = c$ without using our own software?

The solution is a modular test. We take a single-precision prime p and check whether $a \cdot b \equiv c \bmod p$ (read "$a \cdot b$ and c are **congruent modulo** p"), which means that $a \cdot b - c$ is divisible by p, or equivalently, $a \cdot b$ and c have the same remainder on division by p. By (1) below, it is sufficient for this purpose to compute the remainders $a^* = a \text{ rem } p$, $b^* = b \text{ rem } p$, $c^* = c \text{ rem } p$ and check whether $a^* \cdot b^* \equiv c^* \bmod p$, since $a \cdot b \equiv a^* \cdot b^* \bmod p$. If this fails to hold, then clearly somewhere there is an error. How reliable is this test?

Of course, it can happen that $ab \neq c$ and $a^*b^* \equiv c^* \bmod p$. This happens if and only if $ab - c \neq 0$ is divisible by p. If each of our primes is at least 2^{63}, then the product of k of them at least $2^{63 \cdot k}$. Now $|ab - c|$ is a number with not more than 20000 words, and hence divisible by at most $\log_{2^{63}}(2^{64 \cdot 20000}) \leq 20318$ different primes. If we have a data base of 40636 single-precision primes and choose p at random from these, then the test will fail to detect a software error with probability at most $1/2$ (assuming that the test itself is correctly implemented). There is an abundant supply of primes: over $2 \cdot 10^{17}$ 64-bit primes, and more than 90 million 32-bit primes, by the prime number theorem 18.7 (see also Exercise 18.18).

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By standard tricks, such as rerunning the test or choosing a larger data base, this probability can be made arbitrarily small.

The technique can also be used to test equalities like $f \cdot g = h$ for polynomials f, g, h, by substituting a random value, or $A \cdot B = C$ for matrices A, B, C, by evaluating at a random vector.

This **fingerprinting** method can even be applied to problems outside the algebraic realm, by "arithmetizing" combinatorial problems. Suppose that one maintains a large data base in North America and a mirror image in Europe, by performing all updates on both. Each night, one wants to check whether they indeed are identical. Sending the whole data base would take too long. So one considers the data base as a string of words, many gigabytes long, and the (large) number a whose 2^{64}-ary representation this is. Then one chooses a prime p, computes a rem p and sends this to the mirror site. The corresponding calculation is performed on the other data base, and the two results are compared. If they disagree, then the two data bases differ. If they agree, then probably the two data bases are identical, provided p was chosen appropriately. This can be set up so that the size of the transmitted message is only logarithmic in the size of the data bases. Exercise 4.3 asks you to apply this method to the more general problem of string matching.

Division with remainder of a large number by a small number is easy (Exercise 4.1), and you are familiar with one particularly simple example: the remainder of a number modulo 9 (or 3) equals the remainder of the sum of its decimal digits. In particular, the number is divisible by 9 (or 3) if and only if this sum is. Why does this work? Let $a = \sum_{0 \le i < l} a_i \cdot 10^i$ be the decimal representation of $a \in \mathbb{N}$. Since $10 \equiv 1 \bmod 9$ (and $\bmod 3$), we have $a \equiv \sum_{0 \le i < l} a_i \cdot 1^i = \sum_{0 \le i < l} a_i \bmod 9$ (and $\bmod 3$) (see Exercise 4.4 for remainders modulo 11).

Computing with remainders of arithmetic expressions modulo some nonzero integer is called **modular arithmetic**. Given an expression e involving integers and arithmetic operations $+, -, \cdot$, we can compute e modulo some number m very efficiently by first reducing all integers modulo m and then, step by step, performing an arithmetic operation in $\mathbb{Z}$ and immediately reducing the result modulo m again, as we have done in the examples above. Here is another one:

$$e = 20 \cdot (-89) + 32 \equiv 6 \cdot 2 + 4 \equiv 12 + 4 \equiv 5 + 4 \equiv 9 \equiv 2 \bmod 7.$$

In this way, the intermediate results never exceed m^2. The basic rules for computing with congruences are

$$a \equiv b \bmod m \Longrightarrow a * c \equiv b * c \bmod m, \tag{1}$$

where $a, b, c \in \mathbb{Z}$ and $*$ is one of $+, -, \cdot$. Using these rules inductively, we may replace numbers by congruent ones in any arithmetic expression involving additions, subtractions, and multiplications without changing the result modulo m. In other

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words, we may manipulate congruences as we are used to manipulating equations, with one important exception: we have to be careful about cancellation and division. For example, we have $0 \cdot 2 \equiv 2 \cdot 2 \equiv 0 \bmod 4$, but $0 \not\equiv 2 \bmod 4$.

We can also perform modular arithmetic with polynomials over a ring R. Here the simplest nontrivial modulus is a linear polynomial $x-u$ with $u \in R$. For any polynomial $f \in R[x]$, the polynomial $f(x)-f(u)$ has u as a zero and hence is divisible by $x-u$. If we let $q=(f(x)-f(u))/(x-u)$, then $f=q\cdot(x-u)+f(u)$, and since $f(u)$ is a constant polynomial, its degree is less than $1=\deg(x-u)$. By the uniqueness of the remainder, $f(u)$ is the remainder of f on division by $x-u$, that is, $(f \text{ rem } (x-u))=f(u)$ and $f \equiv f(u) \bmod (x-u)$. Thus calculating modulo $x-u$ is the same as evaluating at u.

This can be used to check equality of polynomial expressions, as follows. Suppose that we want to know whether the two polynomials $f=(x-1)(x-2)\cdot(x-3)(x-4)(x-5)+1$ and $g=x^5-15x^4+85x^3-223x^2+274x-119$ are equal. To check this, we evaluate them at some number u. Let us try $u=0$. This is easy for g, since $g \text{ rem } x=g(0)=-119$ is just its constant coefficient (this is analogous to computing a rem 10 for an integer a given in decimal representation), and $f \text{ rem } x=f(0)=(-1)(-2)(-3)(-4)(-5)+1=-120+1=-119$, so this does not hinder f from being equal to g. If we take $u=1$, then computing $f \text{ rem } (x-1)=f(1)=1$ is easy, since the first summand contains the linear factor $x-1$. Now $g \text{ rem } (x-1)=g(1)=1-15+85-223+274-119=3$ (this is like computing the sum of the decimal digits of an integer a; here $x \equiv 1 \bmod (x-1)$ and hence $x^i \equiv 1^i = 1 \bmod (x-1)$ for $i \in \mathbb{N}$), and we conclude that $f \neq g$. In fact, we would have even found this out if we had done the whole computation modulo the polynomial $x-1$ *and* modulo the integer 5, since $g(1) \equiv 1-0+0-3+4-4 \equiv 3 \not\equiv 1 = f(1) \bmod 5$.

If our polynomial modulus is not linear, then things are a bit more complicated. For example, if $m=x^2-x-1$, then the remainder of a polynomial f on division by m is a unique polynomial of degree at most one. For $f=x^3$, we have $x^3 \equiv 2x+1 \bmod m$, $x^2+2x \equiv 3x+1 \bmod m$, and hence

$$\begin{aligned}(x^3+1)(x^2+2x)-x^3 &\equiv ((2x+1)+1)(3x+1)-(2x+1)\\ &= (6x^2+8x+2)-(2x+1) \equiv (14x+8)-(2x+1)\\ &= 12x+7 \bmod x^2-x-1,\end{aligned}$$

since $6x^2+8x+2 \equiv 14x+8 \bmod m$.

The mathematical concept behind modular arithmetic is the **residue class ring**. If R is a ring (say $R=\mathbb{Z}$ or $R=F[x]$) and $m \in R$, then $\langle m \rangle = mR = \{mr : r \in R\}$, the set of all multiples of m, is the **ideal** generated by m, and the residue class ring $R/mR = R/\langle m \rangle = \{f \bmod m : f \in R\}$ consists of all **residue classes** $f \bmod m = \{f+mr : r \in R\}$ for $f \in R$. (The notation $f \bmod m$ for the residue class is somewhat unfortunate since it may lead to confusion with congruences like $f \equiv g \bmod m$,

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but it is nevertheless widely used.) For example, if we take $R = \mathbb{Z}$ and $m = 5$, then $3 \bmod 5 = \{\ldots, -12, -7, -2, 3, 8, 13, \ldots\}$ is one residue class of the residue class ring $\mathbb{Z}/5\mathbb{Z} = \{0 \bmod 5, 1 \bmod 5, 2 \bmod 5, 3 \bmod 5, 4 \bmod 5\}$. The elements of $3 \bmod 5$ can be characterized as those integers having remainder 3 on division by 5, and any two of them are congruent modulo 5. We also write $\mathbb{Z}_m$ for $\mathbb{Z}/m\mathbb{Z}$. If $R = \mathbb{Q}[x]$ and $m = x^2 - x - 1$, then $12x + 7 \bmod m$ consists of all polynomials having remainder $12x + 7$ on division by $x^2 - x - 1$, and any two of them are congruent modulo $x^2 - x - 1$.

Computationally, we work with a **system of representatives** (Section 25.2), so that the residue class $f \bmod m$ is represented by its smallest nonnegative element in the case $R = \mathbb{Z}$ and by its unique polynomial of least degree in the case $R = F[x]$; in both cases, the representative equals $f \operatorname{rem} m$. In the integer case, we will often ignore the somewhat cumbersome distinction between residue classes and representatives. For example, we identify $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ and compute with the elements of the latter modulo 5. (The symmetric system of representatives of least absolute value is also useful, for example, $\{-2, -1, 0, 1, 2\}$ for $\mathbb{Z}/5\mathbb{Z}$.) In the polynomial case, however, we will keep this distinction.

By applying the canonical ring homomorphism $a \longmapsto a \bmod m$, congruences in R (like $4 \cdot 2 \equiv 3 \bmod 5$ in $\mathbb{Z}$ or $x^2 \cdot x \equiv 4 \bmod (x^3 - 4)$ in $\mathbb{Q}[x]$) become equalities in R/mR ($4 \cdot 2 = 3$ in $\mathbb{Z}_5$ or $\alpha^2 \cdot \alpha = 4$ in $\mathbb{Q}[x]/\langle x^3 - 4\rangle$, where $\alpha = x \bmod x^3 - 4$), and the rules (1) guarantee that we may work freely with representatives in these equations as long as we stay within the same residue class.

As indicated above, we will often perform computations modulo polynomials *and* modulo integers, that is, in "double" residue class rings of the form $\mathbb{Z}_m[x]/\langle f\rangle$ for some polynomial $f \in \mathbb{Z}_m[x]$. Then we have two levels of computation: coefficient operations, which are performed modulo m, and polynomial operations, which are performed modulo f and are defined in terms of coefficient operations. For example,

$$(4x + 1)(3x^2 + 2x) = 12x^3 + 11x^2 + 2x \quad \text{in } \mathbb{Z}[x],$$

and hence

$$(4x + 1)(3x^2 + 2x) = 2x^3 + x^2 + 2x \quad \text{in } \mathbb{Z}_5[x].$$

Now division with remainder yields

$$2x^3 + x^2 + 2x = 2 \cdot (x^3 + 4x) + x^2 - 6x \quad \text{in } \mathbb{Z}[x],$$
$$2x^3 + x^2 + 2x = 2 \cdot (x^3 + 4x) + x^2 + 4x \quad \text{in } \mathbb{Z}_5[x].$$

Thus

$$(4x + 1)(3x^2 + 2x) \equiv x^2 + 4x \bmod (x^3 + 4x) \quad \text{in } \mathbb{Z}_5[x],$$

or equivalently,

$$(4\alpha + 1)(3\alpha^2 + 2\alpha) = \alpha^2 + 4\alpha \quad \text{in } \mathbb{Z}_5[x]/\langle x^3 + 4x\rangle,$$

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where $\alpha = (x \bmod (x^3+4x)) \in \mathbb{Z}_5[x]/\langle x^3+4x\rangle$. We have done the calculations in detail in this example to illustrate the principle; later we will suppress the details.

4.2. Modular inverses via Euclid

We have seen in the previous section how modular addition and multiplication works. What about inversion and division? Do expressions like $a^{-1} \bmod m$ and $a/b \bmod m$ make sense, and if so, how can we compute their value? The following theorem gives an answer when the underlying ring R is a Euclidean domain.

THEOREM 4.1.
Let R be a Euclidean domain, $a,m \in R$, and $S = R/mR$. Then $a \bmod m \in S$ is a unit if and only if $\gcd(a,m) = 1$. In this case, the modular inverse of $a \bmod m$ can be computed by means of the Extended Euclidean Algorithm.

PROOF. We have

$$\begin{aligned} a \text{ is invertible modulo } m &\iff \exists s \in R \quad sa \equiv 1 \bmod m \\ &\iff \exists s,t \in R \quad sa + tm = 1 \implies \gcd(a,m) = 1. \end{aligned}$$

If, on the other hand, $\gcd(a,m) = 1$, then the Extended Euclidean Algorithm provides such $s,t \in R$. □

EXAMPLE 4.2. We let $R = \mathbb{Z}$, $m = 29$, and $a = 12$. Then $\gcd(a,m) = 1$, and the Extended Euclidean Algorithm computes $5 \cdot 29 + (-12) \cdot 12 = 1$. Thus $(-12) \cdot 12 \equiv 17 \cdot 12 \equiv 1 \bmod 29$, and hence 17 is the inverse of 12 modulo 29. ◇

EXAMPLE 4.3. Let $R = \mathbb{Q}[x]$, $m = x^3 - x + 2$, and $a = x^2$. The last row in the Extended Euclidean Algorithm for m and a is

$$\Big(\frac{1}{4}x + \frac{1}{2}\Big)(x^3 - x + 2) + \Big(-\frac{1}{4}x^2 - \frac{1}{2}x + \frac{1}{4}\Big)x^2 = 1,$$

and $(-x^2 - 2x + 1)/4$ is the inverse of x^2 modulo $x^3 - x + 2$. ◇

A consequence of Theorem 4.1 is that if $S = \mathbb{Z}_p$ for a prime $p \in \mathbb{N}$ or $S = F[x]/\langle f\rangle$ for a field F and an **irreducible polynomial** $f \in F[x]$, so that f has no nonconstant factor of degree less than $\deg f$, then any element of $S \backslash \{0\}$ is a unit, so that S is a field. We will use the notation $\mathbb{F}_p$ for the finite field $\mathbb{Z}_p$ with p elements throughout the rest of this book. More generally, if $f \in \mathbb{Z}_p[x] = \mathbb{F}_p[x]$ is an irreducible polynomial of degree n, then $\mathbb{F}_p[x]/\langle f\rangle$ is a **finite field** $\mathbb{F}_q$ with $q = p^n$ elements; Section 25.4 gives some basic properties of finite fields. In fact,

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this construction works for any prime power q, namely, there exist irreducible polynomials in $\mathbb{F}_q[x]$ of any degree, and any two irreducible polynomials of the same degree lead to isomorphic fields.

EXAMPLE 4.4. Let $R = \mathbb{F}_5[x]$, $f = x^3 - x + 2$, and $a = x^2$. Then f has no zeroes in $\mathbb{F}_5$, and hence is irreducible since its degree is 3 (Exercise 4.33). Thus $\mathbb{F}_{125} = \mathbb{F}_5[x]/\langle f \rangle$ is a field. The last row in the Extended Euclidean Algorithm for f and a in $\mathbb{F}_5[x]$ is

$$(-x-2)(x^3 - x + 2) + (x^2 + 2x - 1)x^2 = 1.$$

Hence $x^2 + 2x - 1$ is the inverse of x^2 modulo f. If we write $\alpha = x \bmod f$ for short, then $\alpha^2 + 2\alpha - 1 = (\alpha^2)^{-1}$ in $\mathbb{F}_{125}$. $\diamond$

The following lemma says that computing modulo an irreducible polynomial f is the same as "adjoining a root of f".

LEMMA 4.5. *Let F be a field, $f \in F[x]$ a monic nonconstant irreducible polynomial, and $K = F[x]/\langle f \rangle$. Then K is an extension field of F, and $f(\alpha) = 0$ for $\alpha = (x \bmod f) \in K$.*

PROOF. K is a field according to Theorem 4.1, and

$$f(\alpha) = f(x \bmod f) = (f(x) \bmod f) = 0. \square$$

The following two corollaries are a consequence of the analyses in Chapter 2 and Theorems 3.13, 3.16, and 4.1.

COROLLARY 4.6.
Let F be a field, and $f \in F[x]$ of degree $n \in \mathbb{N}$. One arithmetic operation in $F[x]/\langle f \rangle$, that is, addition, multiplication, or division by an invertible element, can be done using $O(n^2)$ arithmetic operations in F. More precisely, we have at most $4n^2 + O(n)$ operations for a multiplication modulo f and at most $6n^2 + O(n)$ operations for an inversion modulo f.

COROLLARY 4.7.
One arithmetic operation in $\mathbb{Z}_m$, where $m \in \mathbb{N}_{>0}$ and $n = \lambda(m) = \lfloor (\log_2 m)/64 \rfloor + 1$ is the length of m in the standard representation, can be done using $O(n^2)$ word operations.

If m is a positive integer, then the set $\mathbb{Z}_m^\times$ of those elements of $\mathbb{Z}_m$ that have multiplicative inverses is a group under multiplication, the **group of units** of the ring $\mathbb{Z}_m$

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(Sections 25.1 and 25.2). Theorem 4.1 says that $\mathbb{Z}_m^\times = \{a \bmod m : \gcd(a,m) = 1\}$. **Euler's totient function** $\varphi : \mathbb{N}_{>0} \longrightarrow \mathbb{N}_{>0}$ counts the number of elements in $\mathbb{Z}_m^\times$:

$$\varphi(m) = \#\mathbb{Z}_m^\times = \#\{0 \le a < m : \gcd(a,m) = 1\}.$$

By convention, $\varphi(1) = 1$. If $m = p$ is a prime, then all nonzero elements of $\mathbb{Z}_p = \mathbb{F}_p$ are invertible, and hence $\varphi(p) = p - 1$. More generally, if $m = p^e$ is a prime power, with $e \ge 1$, then Theorem 4.1 implies that $a \bmod p^e$ is invertible if and only if p does not divide a. Thus there are precisely p^{e-1} nonunits in $\mathbb{Z}_{p^e}$, namely the elements $bp \bmod p^e$ with $0 \le b < p^{e-1}$, and we obtain

$$\varphi(p^e) = p^e - p^{e-1} = (p-1)p^{e-1}. \tag{2}$$

In Section 5.4, we will derive a formula for $\varphi(m)$ when m is arbitrary from the Chinese Remainder Theorem. Exercise 4.19 discusses the analog of Euler's totient function for polynomials over a finite field.

4.3. Repeated squaring

An important tool for modular exponentiation is **repeated squaring** (or square and multiply). In fact, this technique works in any set with an associative multiplication, but we will mainly use it in residue class rings.

ALGORITHM 4.8 Repeated squaring.
Input: $a \in R$, where R is a ring with 1, and $n \in \mathbb{N}_{>0}$.
Output: $a^n \in R$.

1. { binary representation of n }
   write $n = 2^k + n_{k-1} \cdot 2^{k-1} + \cdots + n_1 \cdot 2 + n_0$, with all $n_i \in \{0,1\}$
   $b_k \longleftarrow a$

2. **for** $i = k-1, k-2, \ldots, 0$ **do**
   if $n_i = 1$ **then** $b_i \longleftarrow b_{i+1}^2 a$ **else** $b_i \longleftarrow b_{i+1}^2$

3. **return** b_0

Correctness follows easily from the invariant $b_i = a^{\lfloor n/2^i \rfloor}$. This procedure uses $\lfloor \log n \rfloor$ squarings plus $w(n) - 1 \le \lfloor \log n \rfloor$ multiplications in R, where log is the binary logarithm and $w(n)$ is the **Hamming weight** of the binary representation of n (Chapter 7), that is, the number of ones in it. Thus the total cost is at most $2 \log n$ multiplications. For example, the binary representation of 13 is $1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2 + 1$ and has Hamming weight 3. Thus a^{13} would be computed as

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$((a^2 \cdot a)^2)^2 \cdot a$, using three squarings and two multiplications. If $R = \mathbb{Z}_{17} = \mathbb{Z}/\langle 17\rangle$ and $a = 8 \bmod 17$, then we compute $8^{13} \bmod 17$ as

$$\begin{aligned} 8^{13} &\equiv ((8^2 \cdot 8)^2)^2 \cdot 8 \equiv ((-4 \cdot 8)^2)^2 \cdot 8 \\ &\equiv (2^2)^2 \cdot 8 = 4^2 \cdot 8 \equiv -1 \cdot 8 = -8 \bmod 17, \end{aligned}$$

which is much faster than first evaluating $8^{13} = 549\,755\,813\,888$ and then dividing by 17 with remainder. This method was already used by Euler (1761). He calculated $7^{160} \bmod 641$ by computing $7^2, 7^4, 7^8, 7^{16}, 7^{32}, 7^{64}, 7^{128}, 7^{160} = 7^{128} \cdot 7^{32}$, reducing modulo 641 after each step. (He also listed, unnecessarily, 7^3.) As another example, starting from $2^{2^3} = 2^8 = 256$, we only need two squarings modulo $5 \cdot 2^7 + 1 = 641$ to calculate $((2^8)^2)^2 = 2^{2^5} \equiv -1 \bmod 641$. This shows that 641 divides the fifth Fermat number $F_5 = 2^{2^5} + 1$, as discovered by Euler (1732/33); see Sections 18.2 and 19.1. Even if we were given the 10-digit number $2^{2^5} + 1 = 4\,294\,967\,297$, it would seem more laborious to divide it by 641 with remainder rather than to use modular repeated squaring.

There are clever ways of reducing the cost to $(1 + o(1)) \log n$ multiplications (Exercise 4.21), where $o(1)$ goes to zero as n gets large. On the other hand, starting with an indeterminate x and using d multiplications or additions, one can only compute polynomials of degree at most 2^d, and thus $\lceil \log n \rceil$ multiplications are indeed necessary to obtain x^n. However, when x is not an indeterminate but from a well-structured domain, one can sometimes exploit that structure for faster exponentiation algorithms. We will see an example in the iterated Frobenius algorithm of Section 14.7. Particularly important for cryptographic applications are methods based on normal bases and Gauß periods in finite fields.

4.4. Modular inverses via Fermat

Let $p \in \mathbb{N}$ be a prime and $a, b \in \mathbb{Z}$. Then the binomial theorem implies that

$$(a+b)^p = \sum_{0 \le i \le p} \binom{p}{i} a^{p-i} b^i = a^p + p a^{p-1} b + \frac{p(p-1)}{2} a^{p-2} b^2 + \cdots + b^p.$$

If $0 < i < p$, then the binomial coefficient

$$\binom{p}{i} = \frac{p!}{i!(p-i)!}$$

is divisible by p, since the numerator is and the denominator is not. Thus we have the surprising consequence that $(a+b)^p \equiv a^p + b^p \bmod p$, or equivalently, $(\alpha + \beta)^p = \alpha^p + \beta^p$ in $\mathbb{Z}_p$, with $\alpha = a \bmod p$ and $\beta = b \bmod p$ ("the Freshman's dream"). More generally, we obtain by induction on i that

$$(a+b)^{p^i} \equiv a^{p^i} + b^{p^i} \bmod p \text{ for all } i \in \mathbb{N}. \tag{3}$$

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Using this property, we obtain a pretty, elementary proof of the following famous number-theoretic theorem, which—in a more general form—will have many applications in factoring polynomials and primality testing (Chapters 14 and 18).

THEOREM 4.9 *Fermat's little theorem.*
If $p \in \mathbb{N}$ is prime and $a \in \mathbb{Z}$, then $a^p \equiv a \bmod p$ and, if $p \nmid a$, then $a^{p-1} \equiv 1 \bmod p$.

PROOF. It is sufficient to prove the claim for $a \in \{0, \ldots, p-1\}$, which we do by induction on a. The case $a = 0$ is trivial, and for $a > 1$, we have

$$a^p = ((a-1)+1)^p \equiv (a-1)^p + 1^p \equiv (a-1) + 1 = a \bmod p,$$

by (3) and the induction hypothesis. If $a \neq 0$, then a is invertible modulo p, by Theorem 4.1, and the claim follows by multiplying with $a^{-1} \bmod p$. □

Fermat's theorem, together with repeated squaring, gives us an alternative to compute inverses in $\mathbb{Z}_p$: since $a^{p-2}a = a^{p-1} \equiv 1 \bmod p$ for $a \in \mathbb{Z}$ with $p \nmid a$, we have $a^{-1} \equiv a^{p-2} \bmod p$. This may be computed using $O(\log p)$ multiplications and squarings modulo p, using the repeated squaring algorithm 4.8, in total $O(\log^3 p)$ word operations, by Corollary 4.7. This is cubic in the input size, while the time for modular inversion via Euclid's algorithm is only quadratic, but it may be useful for hand calculations.

4.5. Linear Diophantine equations

Another application of the Extended Euclidean Algorithm is the solution of **linear Diophantine equations**. Let $a, f, g \in \mathbb{Z}$ be given, and suppose that we are looking for all *integral* solutions $s, t \in \mathbb{Z}$ of the equation

$$sf + tg = a. \tag{4}$$

The set of all real solutions of (4) is a line in the plane $\mathbb{R}^2$, a one-dimensional object, that can be written as a sum $v + U$ of a particular solution $v \in \mathbb{R}^2$ and the set U of all solutions of the homogeneous equation

$$sf + tg = 0. \tag{5}$$

The following lemma says that this is also true for the set of integral solutions. Moreover, we can decide whether (4) is solvable over $\mathbb{Z}$, and if so, compute all solutions with the Extended Euclidean Algorithm. Since the proof is the same, we state the result for arbitrary Euclidean domains.

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THEOREM 4.10.
Let R be a Euclidean domain, $a,f,g \in R$, and $h = \gcd(f,g)$.

(i) Equation (4) has a solution $(s,t) \in R^2$ if and only if h divides a.

(ii) If $h \neq 0$ and $(s^*,t^*) \in R^2$ is a solution of (4), then the set of all solutions is $(s^*,t^*)+U$, where

$$U = R \cdot \left(\frac{g}{h}, -\frac{f}{h}\right) \subseteq R^2$$

is the set of all solutions to the homogeneous equation (5).

(iii) If $R = F[x]$ for a field F, $h \neq 0$, (4) is solvable, and $\deg f + \deg g - \deg h > \deg a$, then there is a unique solution $(s,t) \in R^2$ of (4) such that $\deg s < \deg g - \deg h$ and $\deg t < \deg f - \deg h$.

PROOF. (i) If $s,t \in R$ satisfy (4), then $\gcd(f,g)$ divides $sf+tg$ and hence a. Conversely, we assume that $h = \gcd(f,g)$ divides a. The claim is trivial if $h = 0$; otherwise we can compute $s^*,t^* \in R$ such that $s^*f + t^*g = h$, using the Extended Euclidean Algorithm, and $(s,t) = (s^*a/h, t^*a/h)$ solves (4).

(ii) For $(s,t) \in R^2$, we have, since $h \neq 0$ and f/h and g/h are coprime, that

$$(5) \iff \frac{f}{h}s = -\frac{g}{h}t \iff \exists k \in R \quad s = k\frac{g}{h} \text{ and } t = k\frac{-f}{h} \iff (s,t) \in U.$$

Then also

$$(4) \iff f \cdot (s-s^*) + g \cdot (t-t^*) = 0 \iff (s-s^*, t-t^*) \in U.$$

(iii) Let $(s^*,t^*) \in R^2$ be a solution to (4). We divide t^* by f/h with remainder and obtain $q,t \in R$ such that $t^* = qf/h + t$ and $\deg t < \deg f - \deg h$. If we let $s = s^* + qg/h$, then (ii) implies that $(s,t) = (s^*,t^*) + q(g/h, -f/h)$ solves (4). Now both $\deg(tg)$ and $\deg a$ are less than $\deg f + \deg g - \deg h$, and hence so is $\deg s + \deg f = \deg(sf) = \deg(a - tg)$. This proves the existence.

Uniqueness follows from $\deg(s + kg/h) = \deg(kg/h) \geq \deg g - \deg h$ for all $k \in R \setminus \{0\}$, and hence the first component of any solution to (4) different from (s,t) has degree at least $\deg g - \deg h$. □

The situation can be generalized to higher dimensions. The proof of the following theorem is left as Exercise 4.24.

THEOREM 4.11.
Let R be a Euclidean domain, $a, f_1, \ldots, f_n \in R$, with all f_i nonzero, and U the set of all solutions of the homogeneous equation $f_1 s_1 + \cdots + f_n s_n = 0$.

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(i) The linear Diophantine equation

$$f_1 s_1 + \cdots + f_n s_n = a \tag{6}$$

is solvable for $s = (s_1, \ldots, s_n) \in R^n$ if and only if $\gcd(f_1, \ldots, f_n)$ divides a.

(ii) If $s \in R^n$ is a solution of (4), then the set of all integral solutions is $s + U$.

(iii) $U = Ru_2 + \cdots + Ru_n$, where

$$u_i = \Big(\frac{f_i}{h_i}s_{i1}, \frac{f_i}{h_i}s_{i2}, \ldots, \frac{f_i}{h_i}s_{i,i-1}, -\frac{h_{i-1}}{h_i}, 0, \ldots, 0\Big),$$
$$h_{i-1} = \gcd(f_1, \ldots, f_{i-1}) = s_{i1}f_1 + \cdots + s_{i,i-1}f_{i-1}$$

for $2 \le i \le n$ and $h_n = \gcd(f_1, \ldots, f_n)$, with suitably chosen $s_{ij} \in R$.

4.6. Continued fractions and Diophantine approximation

Let R be a Euclidean domain, $r_0, r_1 \in R$, and $q_i, r_i \in R$ for $1 \le i \le \ell$ be the quotients and remainders in the traditional Euclidean Algorithm 3.5 for r_0, r_1. Then eliminating the remainders successively, we obtain

$$\frac{r_0}{r_1} = \frac{q_1 r_1 + r_2}{r_1} = q_1 + \frac{r_2}{r_1} = q_1 + \frac{1}{\frac{r_1}{r_2}} = q_1 + \frac{1}{q_2 + \frac{r_3}{r_2}} = q_1 + \frac{1}{q_2 + \frac{1}{\frac{r_2}{r_3}}}$$
$$= q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \frac{r_4}{r_3}}} = \cdots = q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \frac{1}{\ddots + \frac{1}{q_\ell}}}}.$$

This is called the **continued fraction expansion** of $r_0/r_1 \in K$, where K is the field of fractions of R (Section 25.3). In general, arbitrary elements of R may occur in place of 1 in the "numerators" of a continued fraction, but when all of them are required to be 1 as above, a representation of r_0/r_1 by a continued fraction is computed by the traditional Euclidean Algorithm. To abbreviate, we write $[q_1, \ldots, q_\ell]$ for the continued fraction $q_1 + 1/(q_2 + 1/(\cdots + 1/q_\ell)\cdots)$.

EXAMPLE 4.12. We can rewrite the Euclidean Algorithm from page 45 for $r_0 = 126$ and $r_1 = 35$ as follows.

$$q_1 = \Big\lfloor \frac{r_0}{r_1} \Big\rfloor = \Big\lfloor \frac{126}{35} \Big\rfloor = 3, \quad \frac{r_2}{r_1} = \frac{r_0}{r_1} - q_1 = \frac{21}{35},$$

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$$q_2 = \left\lfloor \frac{r_1}{r_2} \right\rfloor = \left\lfloor \frac{35}{21} \right\rfloor = 1, \quad \frac{r_3}{r_2} = \frac{r_1}{r_2} - q_2 = \frac{14}{21},$$
$$q_3 = \left\lfloor \frac{r_2}{r_3} \right\rfloor = \left\lfloor \frac{21}{14} \right\rfloor = 1, \quad \frac{r_4}{r_3} = \frac{r_2}{r_3} - q_3 = \frac{7}{14},$$
$$q_4 = \left\lfloor \frac{r_3}{r_4} \right\rfloor = \left\lfloor \frac{14}{7} \right\rfloor = 2, \quad \frac{r_5}{r_4} = \frac{r_3}{r_4} - q_4 = 0.$$

Thus the continued fraction expansion of $126/35 \in \mathbb{Q}$ is

$$\frac{126}{35} = \frac{18}{5} = [3,1,1,2] = 3 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{2}}}. \diamond$$

If $R = \mathbb{Z}$, then even an element α of $\mathbb{R}$ may be represented by an (infinite) continued fraction in the sense that its initial segments converge to α with regard to the absolute value. How are these continued fractions computed? The rules from the traditional Extended Euclidean Algorithm 3.6 for computing the quotients $q_1, q_2, \ldots$ for $(r_0, r_1) \in \mathbb{Z}^2$ can be reformulated as follows: set $\alpha_1 = r_0/r_1$, $q_1 = \lfloor \alpha_1 \rfloor$, $\beta_2 = \alpha_1 - q_1$, $\alpha_2 = 1/\beta_2$, and in general $q_i = \lfloor \alpha_i \rfloor$, $\beta_{i+1} = \alpha_i - q_i$, $\alpha_{i+1} = 1/\beta_{i+1}$. Starting with an arbitrary real number α_1, this process defines the continued fraction expansion of α_1, as in Table 4.1. We note that $0 \le \beta_i < 1$; the expansion stops when $\beta_i = 0$, which occurs if and only if α_1 is rational.

For example, when $\alpha_1 = \sqrt{3}$, we obtain

$$q_1 = \lfloor \alpha_1 \rfloor = 1, \quad \beta_2 = \alpha_1 - q_1 = -1 + \sqrt{3},$$
$$\alpha_2 = \frac{1}{\beta_2} = \frac{-1-\sqrt{3}}{(-1+\sqrt{3})(-1-\sqrt{3})} = \frac{-1-\sqrt{3}}{-2} = \frac{1}{2} + \frac{1}{2}\sqrt{3} \approx 1.366,$$
$$q_2 = \lfloor \alpha_2 \rfloor = 1, \quad \beta_3 = \alpha_2 - q_2 = -\frac{1}{2} + \frac{1}{2}\sqrt{3},$$
$$\alpha_3 = \frac{1}{\beta_3} = \frac{2}{-1+\sqrt{3}} = 1 + \sqrt{3} \approx 2.732,$$
$$q_3 = \lfloor \alpha_3 \rfloor = 2, \quad \beta_4 = \alpha_3 - q_3 = -1 + \sqrt{3} = \beta_2,$$

and then the process repeats over and over, yielding $\sqrt{3} = [1, \overline{1,2}, \ldots]$, where the overlined part indicates the period of this ultimately periodic sequence.

Table 4.1 gives examples of continued fraction representations of real numbers. The continued fraction expansion of an irrational number $\alpha \in \mathbb{R}$ is an excellent tool for approximating α by rational numbers with "small" denominator. The substantial theory of Diophantine approximation deals with such questions.

The quality of approximations is best described in the form

$$\left| \alpha - \frac{p}{q} \right| \le \frac{1}{cq^2}. \tag{7}$$

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$r \in \mathbb{R}$	continued fraction expansion of r
$\frac{8}{29}$	$[0,3,1,1,1,2]$
$\sqrt{\frac{8}{29}}$	$[0,1,1,9,2,2,3,2,2,9,1,2,\overline{1,9,2,2,3,2,2,9,1,2},\ldots]$
$\sqrt{3}$	$[1,1,2,\overline{1,2},\ldots]$
$\sqrt[3]{2}$	$[1,3,1,5,1,1,4,1,1,8,1,14,1,10,2,1,4,12,2,3,2,1,3,4,1,1,2,14,3,12,\ldots]$
π	$[3,7,15,1,292,1,1,1,2,1,3,1,14,2,1,1,2,2,2,2,1,84,2,1,1,15,3,13,1,4,\ldots]$
$e = \exp(1)$	$[2,1,2,1,1,4,1,1,6,1,1,8,1,1,10,1,1,12,1,1,14,1,1,16,1,1,18,1,1,20,\ldots]$
$\phi = \frac{1+\sqrt{5}}{2}$	$[1,1,\overline{1},\ldots]$
$\log_2\left(\frac{6}{5}\right)$	$[0,3,1,4,22,4,1,1,13,137,1,1,16,6,176,3,1,1,1,1,3,1,2,1,31,3,1,1,5,\ldots]$

TABLE 4.1: Examples of continued fraction representations of real numbers.

This holds for every continued fraction approximation with $c = 1$. Of three consecutive continued fraction approximations, at least one satisfies (7) with $c = \sqrt{5}$, and for any $c > \sqrt{5}$ there are real numbers α that have only finitely many rational approximations with (7); see Notes 4.6. For comparison, this is about twice as good as decimal fractions, where we restrict q to be a power of 10 and can achieve an approximation error of $1/2q$.

Polynomial analogs of (7) are discussed in Exercises 4.29 and 4.30. The latter shows that every power series f has approximations

$$f \equiv r/t \bmod x^{2n+1}$$

with polynomials r,t such that $\deg r, \deg t \le n$, for infinitely many n. These are just certain Padé approximants (Section 5.9).

i	$[q_1,\ldots,q_i]$	decimal expansion	$\pi - [q_1,\ldots,q_i]$	accuracy
1	3	3.0000000000000000000	0.14159265358979323846	0 digits
2	$\frac{22}{7}$	3.1428571428571428571	−0.00126448926734961868	2 digits
3	$\frac{333}{106}$	3.1415094339622641509	0.00008321962752908752	4 digits
4	$\frac{355}{113}$	3.1415929203539823008	−0.00000026676418906242	6 digits
5	$\frac{103993}{33102}$	3.1415926530119026040	0.00000000057789063439	9 digits

TABLE 4.2: Rational approximations of π.

Table 4.2 shows the rational approximations of π that result from truncating the continued fraction expansion after the ith component for $i = 1,\ldots,5$ and the

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number of correct digits (after the decimal point). Throughout history, people have grappled with practical problems, in architecture, land surveying, astronomy etc., that required "squaring the circle". The Egyptian Rhind Papyrus from about 1650 BC gives the value $(16/9)^2 \approx 3.1604$. Archimedes (278–212 BC) gave a method to approximate π, in principle arbitrarily well, using polygons inscribed and circumscribed to a circle; he proved $3\frac{10}{71} < 25\,344/8069 < \pi < 29\,376/9347 < 3\frac{1}{7}$. The Chinese astronomer Tsu Ch'ung-chih (430–501) determined six decimal digits of π and deduced the approximation $355/113$, which was also found by Adrian Antoniszoon (1527–1607). Lambert (1761) proved that π is irrational, and Lindemann (1882) proved that it is transcendental. An interesting unsolved question asks whether the decimal digits of π are uniformly distributed or even random, in some sense. We do not even know to prove that the digit 1, say, occurs infinitely often!

Table 4.3 shows some steps in our knowledge about the decimal expansion of π. Of the records in the 20th century, we only list some where the number of decimal digits of the number of decimal digits (in the rightmost column) increased. The current world record is an awesome 10 trillion digits, but is unlikely to stand for long.

Archimedes	c. 250 BC	2
Tsu Ch'ung-chih	5th c.	7
Al-Kāshī	1424	14
van Ceulen	1615	35
Machin	1706	100
William Shanks	1853	527
Reitwiesner	1949	2 035
Genuys	1958	10 000
Daniel Shanks & Wrench	1962	100 265
Guilloud & Bouyer	1973	1 001 250
Kanada, Yoshino & Tamura	1982	16 777 206
Kanada, Tamura *et al.*	1987	133 554 400
Kanada & Tamura	1989	1 073 740 000
Kanada & Takahashi	1997	51 539 600 000
Kanada & Takahashi	1999	206 158 430 000
Kanada Laboratory	2002	1 241 100 000 000
Yee & Kondo	2011	10 000 000 000 050

TABLE 4.3: Computations of the decimal digits of π.

William Shanks published a book on his computation of 607 digits, but made an error at the 528th digit. With a modern computer algebra system the first 100000 digits require just a few keystrokes (`evalf[100000](Pi)`, for example, in MAPLE), and bang! there it is on your screen.

The computation of π to many digits is based on deep mathematics and is only possible with the help of fast algorithms for high precision integer and floating point arithmetic, based on the Fast Fourier Transform (Chapter 8) and fast division

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(Chapter 9). It is a good test for computer hardware, which is routinely performed on some supercomputers before shipping. Borwein, Borwein & Bailey (1989) speak from experience:

> *A large-scale calculation of π is entirely unforgiving; it soaks into all parts of the machine and a single bit awry leaves detectable consequences.*

A different large-scale number-theoretic computation discovered the famous Pentium division bug. Nicely (1996) reports in his commendably careful count of twin primes (see page 534 for records): *The first run* [...] *was performed on the only Pentium in the group* [...] *The duplicate run was completed on my wife's 486DX-33 on 4 October 1994. It was immediately clear that their ultra-precision reciprocal sums differed.* [...] *Several days were then spent looking for the culprit: compiler error, memory error, system bus, etc.* [... My] *message and its consequences were spread worldwide over the Internet within days, and eventually Intel admitted that such errors* [...] *were the result of a production flaw in nearly all (over a million) of the Pentium CPUs produced to that time.* [...] *Intel eventually agreed to replace all such chips with corrected versions, at the company's expense. In January 1995, Intel announced (PC Week, 1995) a $475 million accounting charge to cover the cost of the episode.* [...] *As we tell our students—check your work!*

4.7. Calendars

The **tropical year**, to which our calendar adheres, is the period of time between two successive occasions of vernal equinox, the precise point of time in spring when the sun crosses the celestial equator. The length of the tropical year is about $365^d\ 5^h\ 48'\ 45.2''$, or 365.242190 days. (Actually, the exact value is currently diminishing by about 0.53 seconds each century, but this shall not bother us here.)

Since the dawn of civilization, people have used calendars to express the regularities of the moon's rotation around the earth and of the seasons. **Lunar calendars** divide time into months, where originally each month began with new moon. Since the length of a lunar month is between 29 and 30 days, lunar calendars are asynchronous to the year of the seasons. **Solar calendars**, however, ignore the moon phases and try to approximate the year of the seasons as closely as possible.

The early Roman calendar was of a mixed **lunisolar** type. It consisted of originally 10 and later 12 months, and occasionally one extra month was added in order to keep in step with the seasons. The **Julian calendar**, named after Julius Caesar (and which had been invented by the Egyptian Sosigenes), started on 1 January 45 BC. Since the Romans before Caesar had badly neglected the management of the calendar, the year 46 BC, the **annus confusionis**, had 445 days! Caesar used

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the approximation of 365.25 days for the year and introduced one additional 366th **leap day** every four years. Although this approximation is quite close to the exact length of the tropical year, the Julian calendar was fast by about three days every 400 years.

Towards the end of the 16th century, vernal equinox was on 10 March rather than on its "correct" date of 21 March. To rectify this, Pope Gregory XIII introduced the following calendar reform. First, the erroneous calendar gain was fixed by eliminating the 10 days between 4 October and 15 October 1582. Second, the leap year rule was modified by turning those years which are divisible by 100 but not by 400 into normal years; this removed three leap days in 400 years. So, for example, the years 1700, 1800, and 1900 AD were all normal years, but counting the year 2000 AD as normal would be the bug of the millennium. This **Gregorian calendar**, which is essentially still used today, corresponds to an approximation of the tropical year as

$$365 + \frac{1}{4} - \frac{3}{400} = 365\frac{97}{400} = 365.2425$$

days. It is too long by about 26.8 seconds a year.

approximation			difference from tropical year
$365\frac{1}{4}$	$=$	365.25000	$11' 14.8''$
$365\frac{7}{29}$	$=$	365.24137...	$-1' 10.0''$
$365\frac{8}{33}$	$=$	365.24242...	$20.2''$
$365\frac{31}{128}$	$=$	365.24218...	$-0.2''$

TABLE 4.4: Continued fraction approximations of the length of the year.

We obtain other rational approximations to the exact length of the tropical year from its continued fraction expansion $365.242190 = [365, 4, 7, 1, 3, 24, 6, 2, 2]$. Some corresponding approximations are given by Table 4.4. The first is the one that Julius Caesar used. The third approximation is already better than the Gregorian calendar and has a considerably smaller denominator. The **binary calendar** in the last line looks appealing to computer scientists or others familiar with the binary system: a simple rule for its implementation is to have a leap day in year number n if and only if $4 \mid n$ and $128 \nmid n$.

4.8. Musical scales

Another application of continued fractions is in musical theory. A musical **interval** consists of two tones (usually of different pitch, such as G–C), and the term also denotes the sound generated from some instrument playing the two tones one after the other or simultaneously. To each interval corresponds a unique frequency

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ratio of the two tones involved. Table 4.5 lists some common intervals and their frequency ratios.

frequency ratio	name	example
$r_1 = 2:1$	octave	c–C
$r_2 = 3:2$	fifth	G–C
$r_3 = 4:3$	fourth	F–C
$r_4 = 5:4$	major third	E–C
$r_5 = 6:5$	minor third	$E^\flat$–C
$r_6 = 9:8$	whole tone	D–C

TABLE 4.5: Frequency ratios of some musical intervals.

"Calculation" with intervals works as follows: combination of two intervals corresponds to multiplication of their frequency ratios. For example, the octave c–C can be regarded as combining a fifth c–F with a fourth F–C, and in fact the corresponding frequency ratios satisfy $2/1 = (3/2) \cdot (4/3)$.

The origins of musical theory date back to the early Pythagoreans, who were experimenting with the monochord, a stringed instrument with one string of fixed length that can be subdivided into two variable length parts by means of a movable bridge. They found that the tones generated by, for example, one half, two thirds, or three quarters of the string's length, together with the base tone generated by the string in its full length, give intervals that are appealing to the human ear. More generally, they postulated that this is the case if the length ratio (or equivalently, the frequency ratio) of the interval is a quotient of small positive integers.

Pythagorean tuning derives the frequency ratios for all other tones from the frequency ratios $3/2$ for the fifth and $2/1$ for the octave. Diatonic tuning, invented by Didymos, in addition fixes the frequency ratio $5/4$ of the major third. For both tuning schemes, Table 4.6 lists the ratios with respect to the base tone C of the tones in the **diatonic scale** C–D–E–F–G–A–B–c for the key of C major. In the Pythagorean tuning, there are five whole tone steps D–C, E–D, G–F, A–G, and B–A, with frequency ratio $9/8$, plus two half-tone steps F–E and c–B with ratio $254/243$. All integers that occur are powers of 2 or 3. In diatonic tuning, there are three big whole tone steps D–C, G–F, and B–A, with frequency ratio $9/8$, two small whole tone steps E–D and A–G with ratio $10/9$, and two half-tone steps F–E and c–B with ratio $16/15$.

tone	C	D	E	F	G	A	B	c
Pythagorean tuning	1:1	9:8	81:64	4:3	3:2	27:16	243:128	2:1
diatonic tuning	1:1	9:8	5:4	4:3	3:2	5:3	15:8	2:1

TABLE 4.6: Frequency ratios with respect to the base tone C in the diatonic scale.

One disadvantage of both Pythagorean and diatonic tuning is that transposing a piece of music written in one tonal key, say C major, into another tonal key,

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say D major (which corresponds to multiplying all frequencies by the ratio of the interval D–C, thus leaving all frequency ratios unchanged), is not as easily possible with a piano tuned for C major as it is for the human voice. For example, with diatonic tuning the whole tone E–D in C major has frequency ratio $(5/4)/(9/8) = 10/9$ and not $9/8$ as it would be in D major. This led to the invention of the **well-tempered scale**, which dates back to Bartolomé Ramos (1482), who gave suggestions, and to Marin Mersenne (1636), who laid the mathematical foundations. Prominent advocates of the well-tempered scale were the organist and musical scientist Andreas Werckmeister (1691) and the composer Johann Sebastian Bach. *Das Wohltemperierte Klavier*[1] by Bach (1722) promoted, in its 48 preludes and fugues, the idea that all tonal keys can be played on a well-tempered instrument (organ, harpsichord, or piano) without audible dissonances. Today, the well-tempered scale dominates the construction of instruments and the performance of music at least in the Western world.

This scale divides the octave c–C into 12 equal half-tones, corresponding to 12 neighboring keys on the piano (Figure 4.7). The 13 white and black keys between C and c form the **chromatic scale** built from 12 half-tone steps. The 8 white keys alone represent the diatonic scale (for the key of C major) with 5 whole-tone steps D–C, E–D, G–F, A–G, B–A and two half-tone steps F–E and c–B. Why choose this number 12?

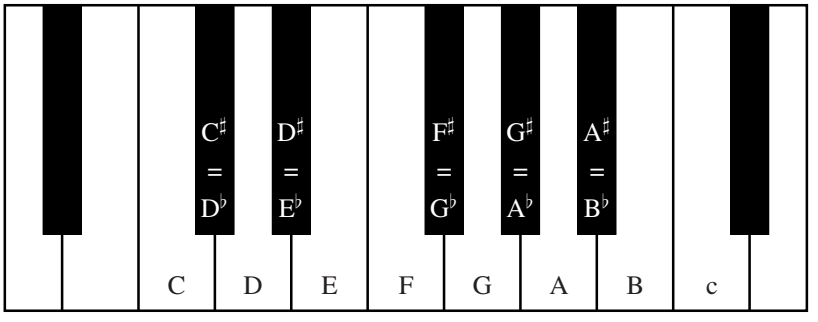


FIGURE 4.7: Part of a piano keyboard.

Suppose that we want to divide the octave into n equal parts, so that one half-tone has frequency ratio $2^{1/n}$, in such a way that each of the pleasant intervals can be reached with an integral number of half-tone steps. This means that for $i = 1, \ldots, 6$, the frequency ratio r_i from Table 4.5 should be close to $2^{d_i/n}$ for some integer d_i; equality would be best, but, for example, there are no integers d, n such that $r_2 = 3/2 = 2^{d/n}$ (Exercise 4.31). Taking logarithms, we have the task of finding $d_i \in \mathbb{N}$ with

$$\left| \log r_i - \frac{d_i}{n} \right| \tag{8}$$

[1] The Well-Tempered Clavier

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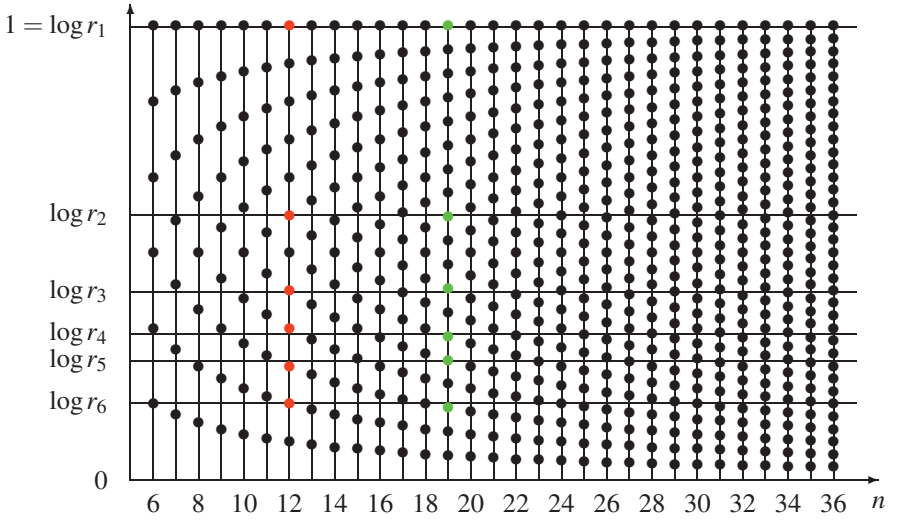
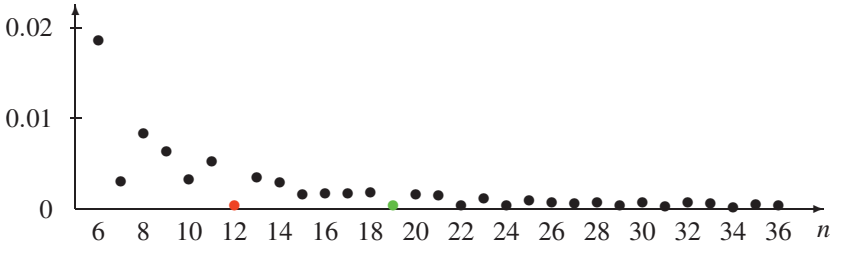


FIGURE 4.8: Diophantine approximations with denominator n to $\log r_1, \ldots, \log r_6$.



Error in the nth approximation

small, where log is the binary logarithm. This is a Diophantine approximation problem and best solved by the continued fraction expansion of $\log r_i$. For $\log r_5 = \log(6/5) = 0.2630344058\ldots$, we find from the continued fraction expansion of Table 4.1 the approximations

$$\begin{aligned} \frac{1}{3} &= 0.3333333333\ldots, \\ \frac{1}{4} &= 0.2500000000, \\ \frac{5}{19} &= 0.2631578947\ldots, \\ \frac{111}{422} &= 0.2630331753\ldots. \end{aligned}$$

The problem of a well-tempered scale consists in finding *one* denominator n such that with *various* numerators d_i the errors (8) are small. This problem of **simultaneous Diophantine approximation** is briefly discussed in Example 17.1 of

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Section 17.3. Here we solve it graphically: in Figure 4.8 on page 87, we have a horizontal line at $\log r_i$, for the six "pleasant intervals" shown, and the dots on the vertical line passing through $n = 6, 7, \ldots, 36$ have distance $0, 1/n, 2/n, \ldots, (n-1)/n, 1$ from the horizontal axis. Inspection reveals a reasonable fit for $n = 12$; the lower diagram depicts the quality of the approximations, defined as the sum over all i of the squares of the distance between $\log r_i$ and its closest neighbor. For $n = 12$, a whole tone is two half-tones, a minor third three, and so on.

Another good fit is at $n = 19$. Then a whole tone is three "third-tones", a minor third is five third-tones, and so on. For example, the tones $E^\flat$ and $D^\sharp$, which are distinguished in music sheets, correspond to the same key on the piano (but are different when played on a violin, say). In a well-tempered 19-tone scale, however, the minor third $E^\flat$–C is five third-tones, while the augmented second $D^\sharp$–C is only four third-tones.

Notes. **4.1.** The fingerprinting technique was invented by Freivalds (1977); an early use was by the Persian philosopher and scientist Avicenna (980–1037), who apparently verified his calculations by checking them modulo 9, as did al-Khwārizmī and al-Kāshī. DeMillo & Lipton (1978) apply it to checking equivalence of two arithmetic circuits. Many other applications in computer science, from universal hashing to probabilistically checkable proofs, are described in Motwani & Raghavan (1995). The general topic of probabilistic algorithms is discussed in Section 6.5, and Section 18.4 presents techniques for finding prime numbers.

4.2. Moore (1896) proved that any finite field is an $\mathbb{F}_q$, as described, for a prime power q, and coined the term "Galois-field"; the notation $GF(q)$ is quite common.

4.3 and 4.4. The term *repeated squaring* seems to have been standard in the early 20th century, since Pocklington (1917) uses it without further explanation (except to say that for his modular problem one divides with remainder by the modulus after each multiplication). Bürgisser, Clausen & Shokrollahi (1997) note that repeated squaring for the computation of a^n is just Horner's rule for the binary representation of the exponent n. Knuth (1998), §4.6.3, discusses at length the topic of addition chains where one tries to minimize the number of multiplications for an exponentiation. We refer to Mullin, Onyszchuk, Vanstone & Wilson (1989), von zur Gathen & Nöcker (1997, 1999), Gao, von zur Gathen & Panario (1998), and Gao, von zur Gathen, Panario & Shoup (2000) for exponentiation in finite fields.

Euler (1732/33) proved that any prime p dividing F_n satisfies $p \equiv 1 \bmod 2^{n+2}$ (Exercise 18.26). He found the factor 641 of F_5 as only the second possibility allowed by his condition. In the same five-page paper, he also states Fermat's little theorem, but saying that he has no proof and *eo autem difficiliorem puto eius demonstrationem esse, quia non est verum, nisi* $n+1$ *sit numerus primus*.[2] Our proof is from his later paper Euler (1736b).

Fermat never communicated a proof of his "little theorem". An unpublished manuscript of Leibniz from 12 September 1680 (and also Leibniz 1697) contains the first proof of Fermat's little theorem; see Mahnke (1912/13), page 38, Vacca (1894), Tropfke (1902), page 62, and Dickson (1919), Chapter III, pages 59–60. Mahnke considers it likely that

[2] I also assume its proof to be rather difficult because it is not true unless $n+1$ is prime.

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Leibniz found the statement of the theorem himself, but it cannot be completely ruled out that he had already read Fermat's *Varia Opera*, published in 1679.

4.5. Generalizing our single linear Diophantine equation, we may consider a system of linear Diophantine equations, that is, a matrix $F \in R^{m\times n}$ and a vector $a \in R^m$, where R is a Euclidean domain, and ask for $s \in R^n$ satisfying $Fs = a$. There are variants of Gaussian elimination that allow only elementary *unimodular* row and column transformations, that is, permutations, multiplications by units of R, and additions of multiples of one row or column to another. They transform the original system into an equivalent one, for example, in **Smith** or in **Hermite normal form**, in which solvability of the system and the set of solutions are easy to determine. For example, the Hermite normal form of a nonsingular square matrix $F \in \mathbb{Z}^{n\times n}$ is the unique lower triangular matrix $H = UF$, where $U \in \mathbb{Z}^{n\times n}$ is unimodular, so that $\deg U = \pm 1$, all diagonal elements of H are positive, and each entry below the diagonal is nonnegative and smaller than the diagonal element in the same column (Exercise 16.7).

Such unimodular transformations correspond to the division steps in the Euclidean Algorithm. For the case of one equation, for example, the Hermite normal form of a one-row matrix $F = (f_1, \ldots, f_n) \in R^{1\times n}$ is $(h, 0, \ldots, 0)$, where $h = \gcd(f_1, \ldots, f_n)$, so that computing the Hermite normal form in this case is the same as computing a gcd. We will encounter a different type of unimodular "Gaussian elimination" for $R = \mathbb{Z}$ in the basis reduction of Chapter 16.

Finally, we may drop the requirement that the equations be linear and ask whether a system of polynomial equations in several variables with coefficients in R has a solution, and if so, look for an explicit description of the set of all solutions. *Hilbert's tenth problem* (see page 587) asks to determine the solvability of a Diophantine equation for $R = \mathbb{Z}$. Against Hilbert's intuition, the question turns out to be undecidable, in the sense of Turing. (See for example Sipser (1997) for the background.) This was proved by Matiyasevich (1970), who showed that any recursively enumerable set $D \subseteq \mathbb{N}$ can be represented as

$$D = \{s_1 \in \mathbb{N} : \exists s_2, \ldots, s_n \in \mathbb{N}\ f(s_1, \ldots, s_n) = 0\}$$

for some $n \in \mathbb{N}$ and some polynomial $f \in \mathbb{Z}[x_1, \ldots, x_n]$. Thus $D \neq \emptyset$ if and only if there is some $s \in \mathbb{N}^n$ such that $f(s) = 0$. By Lagrange's famous theorem, every nonnegative integer can be written as a sum of four squares, and hence $D \neq \emptyset$ if and only if $g(t) = 0$ has an integral solution $t \in \mathbb{Z}^{4n}$, where

$$g = f(y_1^2 + y_2^2 + y_3^2 + y_4^2, \ldots, y_{4n-3}^4 + y_{4n-2}^2 + y_{4n-1}^2 + y_{4n}^2) \in \mathbb{Z}[y_1, \ldots, y_{4n}].$$

Since nonemptiness of an arbitrary recursively enumerable set is undecidable, so is the question whether a Diophantine equation is solvable. Matiyasevich (1993) gives a nice presentation of the proof and its history.

4.6. Cataldi (1513) gives many rational approximations to square roots of integers. He derives from this rules for parading soldiers in square formations; obviously the granting agencies' pressure for "useful applications" is not a recent phenomenon. Leibniz (1701) gives the continued fraction for the golden ratio and the approximation denominators, which are the Fibonacci numbers. Hugenius (1703) treats the approximation of real numbers by fractions in the context of automata with cog wheels. Euler (1737) coined the term *fractiones continuae*, gave the first thirteen entries of the expansion for π, and showed

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that the continued fraction expansion is periodic for a root of an irreducible quadratic polynomial with integer coefficients; Lagrange (1770a) proved the converse. By relating continued fractions to certain differential equations, Euler (1737) derived the expansion for e. In one of his papers on continued fractions, Euler (1762/63) introduces explicit expressions for our numbers s_i and t_i and proves Lemma 3.8 (iv) and (v). Hurwitz (1891) showed the quality $1/\sqrt{5}q^2$ of rational approximations, and the optimality of $c = \sqrt{5}$ in (7). Lagrange (1798) had proven that any "best" rational approximation comes from a continued fraction. Perron's (1929) classic explains the rich and interesting theory of continued fractions in detail; for further reading and references see also Knuth (1998), §4.5.3, and Bombieri & van der Poorten (1995) give an amusing introduction.

Al-Khwārizmī gives in his *Algebra*, written around 825, three values for π: $3\frac{1}{7} \approx 3.1428$, $\sqrt{10} \approx 3.16$, and $62\,832/20\,000 = 3.1416$. Already the Indian mathematician Āryabhaṭa (c. 530) had obtained the latter. The quote at the beginning of this chapter shows that al-Khwārizmī was well aware of the inexact quality of these approximations. The word *algorithm* is derived from al-Khwārizmī's name, indicating his family's origin from the town of Khwarezm, present-day Khiva in Uzbekistan (the old-fashioned "algorism" was a better transliteration than the anagram of *logarithm*). *Algebra* comes from the word al-jabr in the title of his algebra book الكتاب المختصر في حساب الجبر و المقابلة *(al-kitāb al-mukhtaṣar fī ḥisāb al-jabr wa-l-muqābala = The concise book on computing by moving and reducing terms).* جبر (jabara) means "to break" and refers to the technique of moving terms of an equation to the other side so that all resulting terms are positive; he did not allow negative terms. (The Spanish word *algebrista* designates both someone who does algebra or who sets broken bones, as in Cervantes (1615), chapter XV, where one of Don Quixote's companions is healed by an algebrista.) مقابلة (muqābala) is probably the "reduction" of equations by subtracting equal quantities on both sides, and these are his two techniques for solving linear and quadratic equations, the topic of the first part of his book. The influence of al-Khwārizmī's work on Arab, and later medieval European, mathematics was profound.

Al-Kāshī was chief astronomer at Ulugh Beg's court in Samarkand. His *Instructions on the circle's circumference*, written around 1424 (see Luckey 1953), is a remarkable achievement. He presents his calculation of π with great concern about error control, Newton iteration for square roots with just the required precision, and a conversion of his hexagesimal result to decimal notation.

Euler (1736a) introduced the symbols π (§638) and e (§171); they became popular with his *Introductio in analysin infinitorum* (Euler 1748a), but Gauß (1866) still used a different notation. π had been used by Jones (1706) and by Christian Goldbach in 1742. Ludolph van Ceulen (1540–1610) published 20 digits of π in 1596, and his tombstone in the Pieterskerk in Leiden, Holland, recorded 35 digits. It was lost in the 19th century, and on 5 July 2000 a reconstruction was ceremoniously installed. Still, tombstones have not caught on as publication medium. Shanks had calculated 527 digits of π by February 1853, all correct. In March and April 1853, he extended this to 607 digits, incorrectly. The 2011 record by Yee & Kondo took about one year on a home-brew desktop computer with about 60 TB of disk space. This impressive achievement is largely due to ingenious algorithms devised by the Chudnovsky brothers, the Borwein brothers, Richard Brent, and their collaborators. Hilbert's (1893) elegant paper proves the transcendence of e and π on four pages. Berggren, Borwein & Borwein (1997) present a magnificent collection of writings about π. It is a must for π gourmets, and we used their material liberally.

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There are analogs of continued fractions in the polynomial case. If F is a field, then an element α of $F((x^{-1}))$ or $F((x))$, the field of formal Laurent series in x^{-1} or x, respectively, may be represented by an infinite continued fraction whose initial segments converge to α with regard to the degree valuation and the x-adic valuation (Section 9.6), respectively. This is discussed in Exercises 4.29 and 4.30.

4.7. Euler (1737) uses a year of $365^d 5^h 49' 8''$, which is 22.8 seconds longer than our value, to calculate several calendars, including the Julian and Gregorian ones, and $365\frac{8}{33}$. Lagrange (1798), §20, finds several rational approximations to the length of the year, our four among them, via continued fractions based on a year of $365^d 5^h 48' 49''$, 3.8 seconds longer than our assumption. He ends his calculations by admonishing the astronomers to do their homework: *comme les Astronomes sont encore partagés sur la véritable longueur de l'année, nous nous abstiendrons de prononcer sur ce sujet.*[3] We follow Lagrange's modesty.

4.8. Drobisch (1855) first used continued fractions to approximate $\log(3/2)$ by rational numbers in order to divide the octave into equal parts. Ternary continued fractions, studied by Jacobi (1868), can be used to approximate two irrational quantities simultaneously, and Barbour (1948) applied them to the tuning problem by approximating $\log(3/2)$ and $\log(5/4)$.

Actually, perfectly well-tempered instruments are quite rare. In a piano, the thick chords for low notes produce anharmonic overtones ("harmonics"). Octaves usually have a ratio slightly higher than $2:1$, and on violins a fourth in one octave may have a ratio different from the same fourth's ratio in another octave.

Exercises.

4.1 Suppose that on a 64-bit processor you are given a single precision number p with $2^{63} < p < 2^{64}$, and the words making up a positive multiprecision integer a, say of n words. Give an algorithm that computes a rem p in $O(n)$ word operations. You may assume that the processor has a double-by-single-precision division instruction, which takes as input three single precision integers a_0, a_1, p such that $a_1 < p$ and returns single precision integers q, r with $a_1 \cdot 2^{64} + a_0 = qp + r$ and $r < p$. Hint: You have to take care of the leading bit of a.

4.2 Suppose that the two data bases to be compared, as in Section 4.1, are up to 10 GB long and actually different, and that we use single-precision primes p with $2^{63} < p < 2^{64}$. There are at least 10^{17} such primes (Exercise18.18).

(i) Modulo how many primes can they agree at most?

(ii) If we choose our prime p at random, what is the probability that our test gives an incorrect "ok"?

4.3* You are to apply the fingerprinting technique to *string matching*. Given are two strings $x = x_0 x_1 \cdots x_{m-2} x_{m-1}$ and $y = y_0 y_1 \cdots y_{n-2} y_{n-1}$, say consisting of symbols $x_i, y_i \in \{0,1\}$ for all i, of lengths $m < n$, respectively. We want to determine whether x occurs as a substring of y. Let $z_i = y_i y_{i+1} \cdots y_{i+m-1}$ be the substring of length m of y starting at position i, for $0 \le i < n - m$. Thus the task is to determine whether $x = z_i$ for some i.

(i) Describe a simple algorithm that uses $O(mn)$ symbol comparisons.

(ii) Let $a = \sum_{0 \le j < m} x_i 2^i$ and $b_i = \sum_{0 \le j < m} y_{i+j} 2^i$ be the integers whose binary representation (with most significant bit right) is x and z_i, respectively, and $2^{63} < p < 2^{64}$ a single precision prime. Give an algorithm that computes all b_i rem p and compares them to a rem p in $O(n)$ word operations.

[3] since the astronomers have not yet agreed on the true length of the year, we will refrain from making a recommendation on this subject.

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(iii) Any match is certainly found by your algorithm. If $m \le 63k$, $i < n-m$, and p was chosen at random among the at least 10^{17} single precision primes (Exercise 18.18), what is the probability that $x \ne y_i$ and yet $a \equiv b_i \bmod p$ (in terms of k)? What is the probability that some such false match is reported, in terms of k and n? For which k and n is the latter probability below 0.1%?

4.4 Prove that an integer $a = \sum_{0\le i\le l} a_i \cdot 10^i \in \mathbb{N}$ is divisible by 11 if and only if the alternating sum $a_0 - a_1 + a_2 - a_3 \pm \cdots + (-1)^l a_l$ of its decimal digits is.

4.5 Show that for any integer m, congruence $\bmod m$ is an equivalence relation on $\mathbb{Z}$, and prove (1).

4.6 Let $m \in \mathbb{N}_{\ge 1}$ and $f \in \mathbb{Z}_m[x]$ be monic of degree n. Show that the residue class ring $\mathbb{Z}_m[x]/\langle f\rangle$ has m^n elements.

4.7 Is there a $b \in \mathbb{Z}$ such that $6b \equiv 1 \bmod 81$?

4.8 (i) Let $a \in \mathbb{N}$ be such that $0 \le a < 1000$ and the three least significant digits in the decimal representation of $17a$ are 001. What is a?

(ii) Same question when the least significant digits are 209.

4.9 Let $f = x^4 + x^3 + 2x^2 + x + 1$, $g_1 = x$, and $g_2 = x^3 + x$ in $\mathbb{Q}[x]$. Compute polynomials $t_1, t_2 \in \mathbb{Q}[x]$ such that $t_i g_i \equiv 1 \bmod f$ for $i = 1, 2$, if they exist. Is $\mathbb{Q}[x]/\langle f\rangle$ a field?

4.10 Show that the polynomial $f = x^3 + x + 1 \in \mathbb{F}_2[x]$ is irreducible, and compute the inverses of all nonzero elements in $\mathbb{F}_8 = \mathbb{F}_2[x]/\langle f\rangle$ using the Extended Euclidean Algorithm.

4.11 Let $g = x^5 + x + 1 \in \mathbb{F}_2[x]$. For each of the two polynomials

(i) $f = x^3 + x + 1$, (ii) $f = x^3 + 1$

in $\mathbb{F}_2[x]$, do the following. If $f \bmod g$ is a unit in $\mathbb{F}_2[x]/\langle g\rangle$, compute its inverse $h \bmod g$. If $f \bmod g$ is a zero divisor, find a polynomial $h \in \mathbb{F}_2[x]$ of degree less than 5 such that $fh \equiv 0 \bmod g$.

4.12 Prove carefully that $\mathbb{R}[x]/\langle x^2 + 1\rangle$ and $\mathbb{C}$ are isomorphic fields.

4.13 (i) Find a polynomial $f \in \mathbb{F}_7[x]$ of degree less than 4 solving the congruence $(x^2 - 1) \cdot f \equiv x^3 + 2x + 5 \bmod x^4 + 2x^2 + 1$ in $\mathbb{F}_7[x]$.

(ii) Show that the residue class ring $\mathbb{F}_{343} = \mathbb{F}_7[x]/\langle x^3 + x + 1\rangle$ is a field, and compute the inverse of $x^2 \bmod x^3 + x + 1$ in $\mathbb{F}_{343}$.

4.14 (i) Let R be a Euclidean domain and $m, f \in R$. Show that $f \bmod m$ is a zero divisor (see page 227) in $R/\langle m\rangle$ if and only if $\gcd(f, m) \ne 1$ if and only if $f \bmod m$ is not invertible in $R/\langle m\rangle$.

(ii) Give an example of a ring containing nonzero elements that are neither units nor zero divisors.

4.15 Let R be a Euclidean domain and $a, b, c \in R$.

(i) Show that the congruence $ax \equiv b \bmod c$ has a solution $x \in R$ if and only if $g = \gcd(a, c)$ divides b. Prove that in the latter case, the congruence is equivalent to $(a/g)x \equiv (b/g)x \bmod (c/g)$.

(ii) For $R = \mathbb{Z}$ and $a = 5, 6, 7$, determine whether the congruence $ax \equiv 9 \bmod 15$ is solvable, and if so, give all solutions $x \in \{0, \ldots, 14\}$.

4.16* The **degree sequence** of a pair $(f, g) \in (F[x] \setminus \{0\})^2$ of nonzero polynomials over a field F is $(\deg r_0, \deg r_1, \ldots, \deg r_\ell) \in \mathbb{N}^{\ell+1}$, where $r_0, r_1, \ldots, r_\ell$ are the remainders in the Euclidean Algorithm for f and g. How many pairs of polynomials $(f, g) \in (\mathbb{F}_q[x] \setminus \{0\})^2$ over the finite field $\mathbb{F}_q$ with q elements have degree sequence $(4, 3, 1, 0)$? Generalize your answer for arbitrary given degree sequences $(n_0, n_1, \ldots, n_\ell) \in \mathbb{N}^{\ell+1}$ with $n_0 \ge n_1 > \cdots > n_\ell \ge 0$ for $\ell \ge 1$. Hint: Use Exercise 3.18. For all possible degree sequences with $n_0 = 3$ and $n_1 = 2$, list the corresponding pairs of polynomials in $(\mathbb{F}_2[x] \setminus \{0\})^2$.

4.17* This continues Exercise 4.16. Let $\mathbb{F}_q$ be a finite field with q elements and $n, m \in \mathbb{Z}$ with $n \ge m \ge 0$.

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(i) For two disjoint subsets $S, T \subseteq \{0, \ldots, m-1\}$, let $p_{S,T}$ denote the probability that no degree in S and all degrees in T occur in the remainder sequence of the Euclidean Algorithm for two random polynomials in $\mathbb{F}_q[x]$ of degrees n and m, respectively. Prove that $p_{S,T} = q^{-\#S}(1-q^{-1})^{\#T}$.

(ii) For $0 \leq i < m$, let X_i denote the random variable that has $X_i = 1$ if i occurs in the degree sequence of the Euclidean Algorithm for two random polynomials in $\mathbb{F}_q[x]$ of degrees n and m, respectively, and $X_i = 0$ otherwise. Show that $X_0, \ldots, X_{m-1}$ are independent and $\text{prob}(X_i = 0) = 1/q$ for all i.

4.18* Let q be a prime power and $n, m \in \mathbb{N}$ with $n \geq m > 0$. Use Exercise 4.17 to prove the following statements.

(i) The probability that two random polynomials of degree n and m, respectively, in $\mathbb{F}_q[x]$ are coprime is $1 - 1/q$.

(ii) The probability that $n_2 = n_1 - 1$ is $1 - 1/q$.

(iii) The probability that the degree sequence is **normal**, that is, $\ell = m+1$ and $n_{i+1} = n_i - 1$ for $1 \leq i < \ell$, is $(1-1/q)^m \geq 1 - m/q$.

4.19 Let $\mathbb{F}_q$ be a finite field with q elements, $f \in \mathbb{F}_q[x]$ of degree $n > 0$, and $R = \mathbb{F}_q[x]/\langle f \rangle$ the residue class ring modulo f. Then $R^\times$, the set of elements of R that have a multiplicative inverse, is a multiplicative group, and Theorem 4.1 implies that $R^\times = \{g \bmod f : \gcd(f,g) = 1\}$. We denote its cardinality by $\Phi(f) = \#R^\times = \#\{g \in \mathbb{F}_q[x] : \deg g < n \text{ and } \gcd(f,g) = 1\}$.

(i) Prove that $\Phi(f) = q^n - 1$ if f is irreducible.

(ii) Show that $\Phi(f) = (q^d - 1)q^{n-d}$ if f is a power of an irreducible polynomial of degree d.

4.20 Devise a recursive variant of the repeated squaring algorithm 4.8, and also an iterative variant which proceeds from the low order to the high order bits of the binary representation of n. Trace all three algorithms on the computation of a^{45}.

4.21* Give a "repeated fourth powering" algorithm that uses $2\lfloor \log_4 n \rfloor$ squarings and $w_4(n) + 1$ ordinary multiplications, where $w_4(n)$ is the number of nonzero digits in the 4-ary representation of n. Trace your algorithm on the computation of a^{45}. Generalize your algorithm to a "repeated 2^kth powering" algorithm for $k \in \mathbb{N}_{>0}$.

4.22 Compute $15^{-1} \bmod 19$ via Euclid and via Fermat.

4.23 Derive $(a+b)^p \equiv a^p + b^p \bmod p$ for all $a, b \in \mathbb{Z}$ and prime p from Fermat's little theorem.

4.24* (i) Let R be a Euclidean domain, $f_1, \ldots, f_n \in R$, and $h = \gcd(f_1, \ldots, f_n)$. Prove that there exist $s_1, \ldots, s_n \in R$ such that $s_1 f_1 + \cdots + s_n f_n = h$.

(ii) Prove Theorem 4.11.

(iii) Let $l = \text{lcm}(f_1, \ldots, f_n)$. Show that if $R = F[x]$ for a field F, $h \neq 0$, and $\deg a < \deg l$, then there exist $s_1, \ldots, s_n \in R$ solving (6) such that $\deg s_i < \deg l - \deg f_i$.

4.25 Compute integral solutions of the linear Diophantine equations $24s + 33t = 9$ and $6s_1 + 10s_2 + 15s_3 = 7$.

4.26 (i) Expand the rational fractions $14/3$ and $3/14$ into finite continued fractions.

(ii) Convert $[2,1,4]$ and $[0,1,1,100]$ into rational numbers.

4.27 Expand each of the following as infinite continued fractions: $\sqrt{2}$, $\sqrt{2}-1$, $\sqrt{2}/2$, $\sqrt{5}$, $\sqrt{7}$.

4.28 Let R be a Euclidean domain and $q_1, \ldots, q_\ell \in R \setminus 0$. Show that

$$[q_1, \ldots, q_i] = \frac{c_{i+1}(q_1, \ldots, q_i)}{c_{i+1}(0, q_2, \ldots, q_i)}$$

for $1 \leq i \leq \ell$, where c_i is the ith continuant polynomial (Exercise 3.20).

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4.29** This exercise assumes familiarity with valuations and formal Laurent series. Let F be a field. The field $F((x^{-1}))$ of formal Laurent series in x^{-1} consists of expressions of the form

$$g = \sum_{-\infty < j \le m} g_j x^j, \quad g_m, g_{m-1}, \ldots \in F$$

for some $m \in \mathbb{Z}$. We set $\deg g = \max\{j \le m : g_j \ne 0\}$, with the convention that $\deg 0 = -\infty$. This degree function has the usual properties, as the degree of polynomials. In fact, the field $F(x)$ of rational functions is a subfield of $F((x^{-1}))$, and we have $\deg(a/b) = \deg a - \deg b$ for $a, b \in F[x]$.

For a Laurent series g, we obtain the continued fraction $[q_1, q_2, \ldots]$ of g as follows. Set $\alpha_1 = g$, and recursively define $q_i = \lfloor \alpha_i \rfloor \in F[x]$ and $\alpha_{i+1} = 1/(\alpha_i - q_i)$ for $i \in \mathbb{N}_{>0}$. Here, $\lfloor \cdot \rfloor$ extracts the polynomial part, so that $\deg(\alpha_i - q_i) < 0$.

(i) Show that $\deg q_i = \deg \alpha_i$ for all $i \in \mathbb{N}_{>0}$ and $\deg \alpha_i > 0$ if $i \ge 2$.

(ii) Prove that the continued fraction of a rational function $r_0/r_1 \in F((x^{-1}))$, with nonzero $r_0, r_1 \in F[x]$, is finite, and that the q_i are the quotients in the traditional Euclidean Algorithm for r_0, r_1.

(iii) Let $s_0 = t_1 = 1$, $s_1 = t_0 = 0$, and $s_{i+1} = s_{i-1} - q_i s_i$, $t_{i+1} = t_{i-1} - q_i t_i$ for $i \ge 1$, as in the traditional Extended Euclidean Algorithm. Prove that the ith convergent $c_i = [q_1, \ldots, q_{i-1}]$ of g is $c_i = -t_i/s_i$, for all $i \ge 2$.

(iv) Show that $g = -(t_{i-1} - \alpha_i t_i)/(s_{i-1} - \alpha_i s_i)$, and conclude that $\deg(g - c_i) < -2 \deg s_i$ for all $i \ge 2$. Thus if $|h| = 2^{\deg h}$ is the degree valuation of a Laurent series h, then we obtain the analog $|g + t_i/s_i| < |s_i|^{-2}$ of (7).

(v) Now let $i \in \mathbb{N}_{\ge 2}$, $k \ge n = \deg s_i$, $r_0 = \lfloor x^{n+k} g \rfloor$, $r_1 = x^{n+k}$, and $r_i = s_i r_0 + t_i r_1$. Conclude from (iv) that $\deg r_i < k$, and show that $r_i/s_i \equiv r_0 \bmod x^{n+k}$ if $x \nmid s_i$. (In fact, Lemma 11.3 implies that $q_1, \ldots, q_{i-1}$ are the first $i-1$ quotients and r_i is the ith remainder in the traditional Euclidean Algorithm for r_0, r_1.)

4.30** This exercise is an analog of Exercise 4.29, now for Laurent series in x rather than in x^{-1}. Let F be a field. The field $F((x))$ of formal Laurent series in x consists of expressions of the form

$$g = \sum_{m \le j < \infty} g_j x^j, \quad g_m, g_{m+1}, \ldots \in F$$

for some $m \in \mathbb{Z}$. We let $v(g) = \min\{j \ge m : g_j \ne 0\}$, with the convention that $v(0) = \infty$.

For a Laurent series g, we obtain the continued fraction $[q_1, q_2, \ldots]$ of g as follows. Set $\alpha_1 = g$, and recursively define $q_i = \lfloor \alpha_i \rfloor \in F[1/x]$ and $\alpha_{i+1} = 1/(\alpha_i - q_i)$ for $i \in \mathbb{N}_{>0}$. Here, $\lfloor \cdot \rfloor$ extracts the part which is polynomial in $1/x$, so that $v(\alpha_i - q_i) > 0$, or equivalently, $x \mid (\alpha_i - q_i)$.

(i) Prove that $v(fg) = v(f) + v(g)$, $v(1/g) = -v(g)$ if $g \ne 0$, and $v(f+g) \ge \min\{v(f), v(g)\}$, with equality if $v(f) \ne v(g)$, hold for all $f, g \in F((x))$.

(ii) Let $s_0 = t_1 = 1$, $s_1 = t_0 = 0$, and $s_{i+1} = s_{i-1} - q_i s_i$, $t_{i+1} = t_{i-1} - q_i t_i$ for $i \ge 1$, as in the traditional Extended Euclidean Algorithm. Then the s_i, t_i are polynomials in $1/x$. Prove that the ith convergent $c_i = [q_1, \ldots, q_{i-1}]$ of g is $c_i = -t_i/s_i$, for all $i \ge 2$.

(iii) Show that $g = -(t_{i-1} - \alpha_i t_i)/(s_{i-1} - \alpha_i s_i)$, and conclude that $v(g - c_i) > -2v(s_i)$ for all $i \ge 2$. Thus if $|h| = 2^{-v(h)}$ is the x-adic valuation of a Laurent series h, then we obtain the analog $|g + t_i/s_i| < |s_i|^{-2}$ of (7).

(iv) Now assume that $g \in F[[x]]$ is a power series, let $i \in \mathbb{N}_{\ge 2}$, and $n = -v(s_i) \in \mathbb{N}$. Prove that $x^n s_i$ and $x^n t_i$ are polynomials of degree at most n, and conclude that there exist polynomials $s, t \in F[x]$ of degree not more than n such that $x \nmid s$ and $t/s \equiv g \bmod x^{2n+1}$.

4.31 Prove that there do not exist integers d, n such that $3/2 = 2^{d/n}$.

4.32** (Sturm 1835) Let $f \in \mathbb{R}[x]$ have no multiple roots, so that $\gcd(f, f') = 1$, and determine $f_0 = f, f_1 = f', f_2, \ldots, f_\ell, q_1, \ldots, q_\ell \in \mathbb{R}[x]$ similarly as in the traditional Euclidean Algorithm, but according to the modified rule

$$q_i = f_{i-1} \text{ quo } f_i, \quad f_{i+1} = -(f_{i-1} \text{ rem } f_i)$$

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for $1 \le i \le \ell$, with the convention that $f_{\ell+1} = 0$. The difference to the traditional Euclidean Algorithm is the sign of f_{i+1}; this corresponds to taking $\rho_0 = \rho_1 = 1$ and $\rho_i = -1$ for $2 \le i \le \ell$ in the Extended Euclidean Algorithm 3.14. The polynomials $f_0, f_1, \ldots, f_\ell$ form the **Sturm chain** of f. For each $b \in \mathbb{R}$, let $w(b)$ be the number of sign alternations in the sequence $f_0(b), \ldots, f_\ell(b)$. Here, a sign alternation occurs when either $f_i(b) < 0$, $f_{i+1}(b) \ge 0$ or $f_i(b) > 0$, $f_{i+1}(b) \le 0$. Prove **Sturm's theorem**, which says that for all $b, c \in \mathbb{R}$ such that $f(b) \ne 0 \ne f(c)$ and $b < c$, the number of real roots of f in the interval (b, c) is $w(b) - w(c)$. Hint: It is sufficient to prove the theorem for intervals containing at most one zero of all the f_i's. Show that w does not change at a zero of some f_i with $i > 0$, but that w drops by one at a zero of $f_0 = f$.

4.33 Let F be a field.

(i) Show that a polynomial $f \in F[x]$ of degree 2 or 3 is irreducible if and only if it has no roots in F.

(ii) For each of the two fields $F = \mathbb{Q}$ and $F = \mathbb{F}_2$, find a polynomial of degree 4 that is reducible and has no roots in F.

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All is fair in war, love, and mathematics.
Eric Temple Bell (1937)

"Divide *et impera*" is as true in algebra as in statecraft;
but no less true and even more fertile is the maxim "auge *et impera*."
The more to do or to prove, the easier the doing or the proof.
James Joseph Sylvester (1878)

Quando orientur controversiae, non magis disputatione opus erit inter duos philosophos, quam inter duos Computistas. Sufficiet enim, calamos in manus sumere, sedereque ad abacos, et sibi mutuo (accito si placet amico) dicere: *calculemus*.[1]
Gottfried Wilhelm Leibniz (1684)

These [results] must not be taken on trust by the student, but must be worked by his own pen, which must never be out of his hand while engaged in any algebraical process.
Augustus De Morgan (1831)

[1] When controversies arise, there will not be a greater dispute between two philosophers than between two computers. It will be sufficient for them to take pen in hand, sit down with their calculators, and (having summoned a friend, if they like) say to each other: *Let us calculate.*

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